AIML-A05

Privacy-Preserving Federated Learning Framework for Early Brain Tumor Classification Using Multi-Hospital MRI Images

Innovative Ideas to Complete and Strengthen the Project

Pick 2–3 of these as the team's research contribution beyond the base paper. Ideas 1–6 add research depth; 7–12 improve realism and the final demo.

1. **Natural-hospital benchmark:** Treat each source dataset (figshare, SARTAJ, Br35H, BraTS) as a real hospital with its own scanner and label mix, instead of random splits. Results under true site differences are far more convincing than the base paper's setup.
2. **Leakage-corrected re-evaluation:** Show the effect of removing duplicates and enforcing patient-level splits on reported accuracy. If the 98% falls, that is a valuable, honest finding.
3. **FedBN for scanner shift:** Keeping BatchNorm local is a simple change that often helps a lot when hospitals use different MRI machines. Present it as the main method improvement over FedAvg.
4. **Glioma-only hospital study:** Include a BraTS hospital that has only glioma cases and study how such extreme label skew harms FedAvg, and which method (FedProx, FedBN, personalisation) fixes it.
5. **Privacy–utility–communication triangle:** One plot combining ε, accuracy and bytes sent for ResNet50 vs EfficientNetB0 shows which model is best for real hospital networks.
6. **Explanation consistency across hospitals:** Compare Grad-CAM maps of local and global models: does federation make the model focus more consistently on the tumour? Measure overlap with BraTS tumour masks.
7. **Fair contribution and incentives:** Estimate each hospital's contribution (e.g. leave-one-out or Shapley-based) to show smaller hospitals how much they gain by joining.
8. **Client dropout and late joiners:** Simulate hospitals going offline or joining mid-training, as happens in real networks, and report robustness.
9. **Personalised models per hospital:** Fine-tune the global model briefly at each hospital and show local accuracy gains while keeping privacy.
10. **Lightweight edge deployment:** Quantise EfficientNetB0 so a hospital PC without a GPU can participate and diagnose.
11. **Uncertainty-aware referral:** Flag low-confidence MRIs for radiologist review using MC dropout or ensembles of client models.
12. **Federated segmentation extension (future scope):** Extend from classification to tumour segmentation on BraTS with a federated U-Net, following the FeTS initiative.